It is suggested that a Ca: P in the diet should remain 1:1. Since the RDA of phosphorus is 600 mg. During infancy the Ca: P ratio suggested is 1:1.5. Bone mass increases during childhood and adolescence and peak between the age of 20 and 30 years. Peak bone mass is influenced by calcium intake. Other factors like age, sex, genetic factors, hormonal status and exercise also influence. After the age of 30, rate of withdrawal exceeds the rate of deposit, therefore establishing healthy bone mass in childhood and early adulthood is crucial.

ii) Iron

Iron requirement for women is 4mg higher than men. Iron lost through mensuration in women of reproductive age groups are 0.6 mg/day on an average, when spread over the whole month. Those who are taking vegetarian diet should ensure adequate amounts of Vitamin C for enhancing iron absorption.

iii) Zinc

The requirement of zinc is 2.7 mg/d. To get this amount the RDA suggested is 12 mg/d, correcting individual variation. In an Indian population the rate of absorption is 20- 25%.

iv) Vitamins

The requirement of Vitamin A is same for both men and women. The conversion ratio of retinol to ß- carotene is 1:8. The requirement of B vitamins is based on calorie requirement (0.5 mg of thiamine, 0.6 mg of riboflavin and 6.6 mg of niacin per 1000 calories). Hence requirements of B vitamin are higher for moderate and heavy workers. The previous recommendation of 2mg/d of pyridoxine of adults is retained.

An in take of 20 mg of Vitamin C may be sufficient to maintain the ascorbic acid level in the adults. Taken into account that 50% of Vitamin C is lost in cooking, 40 mg is suggested for all adults. FAO/WHO suggested 55µg of vitamin K and 400 IU of vitamin D for adults, where there is minimal exposure to sunlight.

Diet for nutritionally deficient can be improved by

1. Replacing a single cereal with mixed cereals.

2. Inclusion of green leafy vegetables to improve the intake of vitamin A, iron and calcium.

3. Inclusion of inexpensive yellow fruits like papaya or mango and greens to increase vitamin A and C intake.

4. Inclusion of at least 150 ml of milk improves intake of riboflavin, calcium besides improving protein quality of the diet.



According to WHO, a healthy diet for an adult should contain the following:

- Fruits, vegetables, legumes (e.g. lentils, beans), nuts and whole grains (e.g. unprocessed maize, millet, oats, wheat, brown rice) should be included in a days menu.
- At least 400grams (5 portions) of fruits and vegetables a day.
- Unsaturated fats (e.g. found in fish, avocado, nuts, sunflower, ground nut and gingerlly oil) are preferable to saturated fats found in fatty meat, butter, palm and coconut oil,

UNIT_4_ Adult and Old age.indd 73 25-02-2019 15:34:08